Week 1

Case Study 1

Data for Questions 1 to 5

Smart Classroom Ventilation Control

A university building uses a simple automatic ventilation controller for a smart classroom. The controller is intended to be transparent enough for maintenance staff to verify without running a learning algorithm. At the beginning of every class interval, the controller reads four binary indicators:

$$x_1 : \text{carbon dioxide level is high}, \quad x_2 : \text{room occupancy is high},$$

$$x_3 : \text{indoor temperature is above the comfort limit}, \quad x_4 : \text{outside air quality is acceptable}.$$

Each indicator is 1 when the condition is true and 0 otherwise. The ventilation fan is controlled by a McCulloch-Pitts style threshold unit:

$$y = \begin{cases} 1, & \text{if } x_1 + x_2 + x_3 + x_4 \geq \theta, \\ 0, & \text{otherwise.} \end{cases}$$

Here, $y = 1$ means that the fan is switched on. The facilities team wants the fan to switch on only when at least three of the four indicators are active. Each class interval is evaluated independently.

Question 1: Which threshold value implements the desired fan-control rule?

A) $\theta = 1$

B) $\theta = 2$

C) $\theta = 3$

D) $\theta = 4$

Correct Answer: C

Explanation: The fan should switch on when at least three indicators are active, so the threshold must be 3.

Question 2: For input $(1, 0, 1, 1)$, what output does the controller produce?

A) $y = 0$, because only one condition is absent

B) $y = 1$, because three indicators are active

C) $y = 0$, because all four indicators must be active

D) $y = 1$, because the first indicator is active

Correct Answer: B

Explanation: The input contains three 1s. Since $3 \geq \theta$, the output is $y = 1$.

Question 3: How many binary input patterns switch on the fan?

A) 5

B) 4

C) 6

D) 8

Correct Answer: A

Explanation: The fan switches on for exactly three active indicators or four active indicators. The number of patterns is $\binom{4}{3} + \binom{4}{4} = 4 + 1 = 5$.

Question 4: If the threshold is accidentally changed to $\theta = 2$, what happens?

A) The fan switches on less often than intended.

B) The fan switches on when two or more indicators are active.

C) The fan switches on only when all inputs are 1, making activation too strict.

D) The fan never switches on.

Correct Answer: B

Explanation: With $\theta = 2$, any pattern with two, three, or four active indicators produces $y = 1$, so activation becomes more frequent.

Question 5: Which property makes this controller a threshold-logic system?

A) It predicts future classroom occupancy states.

B) It updates weights after every incorrect output during a training process.

C) It requires a hidden layer to make a decision.

D) It compares a summed input value with a fixed threshold.

Correct Answer: D

Explanation: A threshold unit computes a weighted or unweighted sum and compares that sum with a threshold.

Case Study 2

Data for Questions 6 to 10

Scholarship Application Screening

A scholarship office builds a first-stage screening aid. The aid does not make the final award decision; it only flags applications that satisfy a simple rule for immediate review. Each application is represented by two binary features:

$$x_1 : \text{academic score is above the cutoff}, \quad x_2 : \text{financial-need certificate is verified.}$$

The office wants the system to flag an application when at least one of these two conditions is true. A perceptron with weights w_1, w_2, bias b, and decision rule

$$\hat{y} = \begin{cases} 1, & \text{if } w_1 x_1 + w_2 x_2 + b \geq 0, \\ 0, & \text{otherwise} \end{cases}$$

is tested for this screening rule.

Question 6: Which logical behavior is described by the desired screening rule?

A) AND

B) XOR

C) OR

D) XNOR

Correct Answer: C

Explanation: The output is 1 when $x_1 = 1$, or $x_2 = 1$, or both. This is the OR operation.

Question 7: Which parameter setting correctly implements this rule?

A) $w_1 = 1,\ w_2 = 1,\ b = -1.5$

B) $w_1 = 1,\ w_2 = 1,\ b = -0.5$

C) $w_1 = -1,\ w_2 = -1,\ b = 0.5$

D) $w_1 = 1,\ w_2 = -1,\ b = 0$

Correct Answer: B

Explanation: With $w_1 = w_2 = 1$ and $b = -0.5$, the score is negative only for $(0, 0)$. All other input patterns produce a non-negative score.

Question 8: For $w_1 = 1,\ w_2 = 1,\ b = -0.5$, what is the score for input $(0, 1)$?

A) 0.5

B) -0.5

C) 0

D) 1.5

Correct Answer: A

Explanation: The score is $1(0) + 1(1) - 0.5 = 0.5$, so the perceptron outputs 1.

Question 9: What is the decision boundary for the parameter setting $w_1 = 1,\ w_2 = 1,\ b = -0.5$?

A) $x_1 - x_2 - 0.5 = 0$

B) $x_1 + x_2 + 0.5 = 0$

C) $2x_1 + x_2 = 0$

D) $x_1 + x_2 - 0.5 = 0$

Correct Answer: D

Explanation: The boundary is where the perceptron score equals zero: $x_1 + x_2 - 0.5 = 0$.

Question 10: Why can a single perceptron represent this screening rule?

A) The positive and negative cases are linearly separable.

B) The rule requires remembering earlier applications before deciding the next output.

C) The rule has no decision boundary.

D) The output changes randomly for the same input.

Correct Answer: A

Explanation: OR is linearly separable because one straight line can separate $(0, 0)$ from the three positive input patterns.

Case Study 3

Data for Questions 11 to 15

Equipment Handover Record Review

An engineering lab tracks two binary checks before handing over a portable measurement kit:

x_1 : battery pack has been returned, x_2 : sensor module has been returned.

For audit purposes, the lab wants to flag a handover record when exactly one of the two items has been returned. If neither item has been returned or both items have been returned, the record is not flagged. The desired output table is:

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

The lab initially tries to implement this rule with a single perceptron.

Question 11: Which logical behavior does the handover rule represent?

A) AND

B) OR

C) XOR

D) NAND

Correct Answer: C

Explanation: The output is 1 exactly when the two binary inputs are different, which is the XOR rule.

Question 12: What output is assigned to the input pattern $(1, 1)$?

A) $y = 1$, because one item has been returned

B) $y = 0$, because exactly one item has not been returned

C) $y = 1$, because both checks are complete

D) $y = 0$, because both values are the same

Correct Answer: D

Explanation: The rule flags only records with exactly one returned item. For $(1, 1)$, both items have been returned, so $y = 0$.

Question 13: Which geometric property prevents a single perceptron from representing this rule?

A) The positive cases lie on opposite corners of the input square.

B) The inputs are binary rather than real valued, so the rule cannot use a separator.

C) The output has only two possible labels.

D) The rule uses two input variables.

Correct Answer: A

Explanation: The positive cases $(0, 1)$ and $(1, 0)$ are diagonally opposite, so a single straight line cannot separate them from the two negative cases.

Question 14: Which modified lab rule would be linearly separable?

A) Flag exactly one returned item.

B) Flag when the two item statuses are different.

C) Flag only when both items have been returned.

D) Flag when neither or both items have been returned.

Correct Answer: C

Explanation: Flagging only $(1, 1)$ is the AND rule, which is linearly separable.

Question 15: What does linear separability mean in this setting?

A) Every input must have label 1 before a straight boundary can be checked.

B) A line can separate the 1-labeled and 0-labeled inputs.

C) The perceptron must avoid using weights.

D) The input values must be stored in a sequence.

Correct Answer: B

Explanation: Linear separability means that a line can place all positive examples on one side and all negative examples on the other side.

Case Study 4

Data for Questions 16 to 20

Parcel Routing Decision Review

A courier service trains a perceptron to classify parcels into "express" and "standard" routing. Each parcel is described by two numerical features:

$$x_1 : \text{compactness score}, \quad x_2 : \text{priority-label score}.$$

The label is $y = 1$ for express routing and $y = -1$ for standard routing. The perceptron has no bias term and uses the update rule

$$w \leftarrow w + yx$$

whenever $y(w^\top x) \leq 0$. The training samples are processed in the following order:

Parcel	x_1	x_2	y
P_1	2	1	1
P_2	1	2	1
P_3	-1	-1	-1

Training starts with $w = (0, 0)$.

Question 16: What is the weight vector after processing P_1 once?

A) $w = (2, 1)$

B) $w = (1, 2)$

C) $w = (-1, -1)$

D) $w = (0, 0)$

Correct Answer: A

Explanation: Initially $w^\top x = 0$, so $y(w^\top x) = 0$ and an update is made. The new vector is $w = (0, 0) + 1(2, 1) = (2, 1)$.

Question 17: After the update from P_1, what is $w^\top x$ for P_2?

A) 2

B) 3

C) 4

D) 5

Correct Answer: C

Explanation: With $w = (2, 1)$ and $x = (1, 2)$, the score is $2(1) + 1(2) = 4$.

Question 18: With $w = (2, 1)$, is P_3 correctly classified?

A) No, because the raw score $w^\top x$ is greater than zero.

B) No, because $y(w^\top x) < 0$.

C) Yes, because $w^\top x = 0$.

D) Yes, because $y(w^\top x) > 0$.

Correct Answer: D

Explanation: For $P_3 = (-1, -1)$, $w^\top x = -3$. Since $y = -1$, $y(w^\top x) = 3 > 0$, so it is correctly classified.

Question 19: Which condition is needed for the perceptron convergence guarantee?

A) Every feature must be binary.

B) The data must be linearly separable.

C) The learning rate must be zero throughout all training updates.

D) All labels must be positive.

Correct Answer: B

Explanation: The perceptron convergence theorem applies when the training data is linearly separable.

Question 20: Which vector separates the three listed parcels?

A) $w = (-1, 1)$

B) $w = (1, -1)$

C) $w = (-1, -1)$

D) $w = (1, 1)$

Correct Answer: D

Explanation: With $w = (1, 1)$, P_1 and P_2 have positive scores, while P_3 has a negative score, matching the labels.

Week 2

Case Study 1

Data for Questions 1 to 5

Clinic Follow-up Priority Scoring

A clinic uses a simple model to assign a follow-up priority score after remote consultation. The model does not make a diagnosis; it only produces a probability-like score for staff review. Each patient record is represented by two normalized inputs:

$$x_1 : \text{symptom persistence score}, \quad x_2 : \text{abnormal reading score}.$$

The model computes

$$z = 3x_1 + 2x_2 - 2, \qquad \hat{p} = \sigma(z) = \frac{1}{1 + e^{-z}}.$$

A record is sent for priority follow-up when $\hat{p} \geq 0.5$. For the sigmoid function, $\sigma(0) = 0.5$, positive z gives an output above 0.5, and negative z gives an output below 0.5.

Question 1: Why is the sigmoid function suitable for this priority score?

A) It maps a real-valued input to a value between 0 and 1.

B) It removes the need to compute any weighted input value.

C) It always gives a hard class label before thresholding.

D) It forces both input features to have identical weights.

Correct Answer: A

Explanation: A sigmoid maps any real number to a smooth output in $(0, 1)$, which can be used as a probability-like score.

Question 2: For $x_1 = 0.4$ and $x_2 = 0.6$, what is the value of z?

A) 0.2

B) 0.6

C) 0.4

D) 0.8

Correct Answer: C

Explanation: $z = 3(0.4) + 2(0.6) - 2 = 1.2 + 1.2 - 2 = 0.4$.

Question 3: The rule $\hat{p} \geq 0.5$ is equivalent to which condition?

A) $z < 0$

B) $z \geq 0$

C) $z > 1$

D) $z \leq -1$

Correct Answer: B

Explanation: Since $\sigma(0) = 0.5$ and sigmoid is increasing, $\hat{p} \geq 0.5$ exactly when $z \geq 0$.

Question 4: Which input is not sent for priority follow-up?

A) $x_1 = 0.8,\ x_2 = 0.1$

B) $x_1 = 0.2,\ x_2 = 0.8$

C) $x_1 = 0.5,\ x_2 = 0.4$

D) $x_1 = 0.1,\ x_2 = 0.4$

Correct Answer: D

Explanation: For $x_1 = 0.1,\ x_2 = 0.4$, $z = 0.3 + 0.8 - 2 = -0.9$, so $\hat{p} < 0.5$.

Question 5: If the clinic wants fewer records to satisfy $\hat{p} \geq 0.5$, what should happen to the bias term -2?

A) It should become more negative.

B) It should be removed from the model entirely.

C) It should be made equal to $x_1 + x_2$.

D) It should be increased to make every score larger.

Correct Answer: A

Explanation: Making the bias more negative lowers z for the same inputs, so fewer records reach the decision threshold.

Case Study 2

Data for Questions 6 to 10

Delivery Time Prediction Review

A logistics team trains a regression model to predict delivery delay in minutes. For four validation orders, the actual delay y and predicted delay $\hat{y}$ are:

Order	y	$\hat{y}$
1	12	10
2	20	24
3	15	13
4	18	21

The team uses squared error

$$L = (y - \hat{y})^2, \qquad \text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

For gradient-based updates with respect to the prediction,

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y).$$

Question 6: What is the squared error for Order 2?

A) 4

B) 16

C) 9

D) 25

Correct Answer: B

Explanation: For Order 2, the error is $20 - 24 = -4$, so the squared error is 16.

Question 7: What is the mean squared error over the four orders?

A) 7.25

B) 8.00

C) 9.50

D) 8.25

Correct Answer: D

Explanation: The squared errors are $4, 16, 4, 9$. Their sum is 33, so the MSE is $33/4 = 8.25$.

Question 8: Which order contributes the largest squared error?

A) Order 1

B) Order 3

C) Order 2

D) Order 4

Correct Answer: C

Explanation: The largest squared error is 16, which occurs for Order 2.

Question 9: If $y = 20$ and $\hat{y} = 24$, what is $\frac{\partial L}{\partial \hat{y}}$?

A) 8

B) -8

C) 4

D) -4

Correct Answer: A

Explanation: $\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y) = 2(24 - 20) = 8$.

Question 10: A second model predicts $11, 21, 14, 19$. Which model has lower MSE?

A) The original model, because its MSE is 8.25.

B) The second model, because its MSE is 1.00.

C) Both models, because their errors are identical.

D) Neither model, because MSE cannot be computed here.

Correct Answer: B

Explanation: The second model has squared errors $1, 1, 1, 1$, so its MSE is 1.00.

Case Study 3

Data for Questions 11 to 15

Machine Energy Use Prediction

A facilities team monitors configurable machines in a shared workshop where different operating choices lead to different electricity usage patterns. Before deploying a larger model, the team studies one small feedforward network so that engineers can trace every intermediate value by hand. The system uses two real-valued inputs:

$$x_1 : \text{duration score}, \quad x_2 : \text{load score}.$$

The team uses a small feedforward network:

$$h_1 = \sigma(x_1 + x_2 - 1), \quad h_2 = \sigma(2x_1 - x_2),$$

$$\hat{y} = \sigma(3h_1 + 2h_2 - 2).$$

The hidden layer transforms the original features before the final output is computed. This lets the team discuss the order of computation, the role of nonlinear hidden units, and why the final sigmoid produces a bounded score rather than an unrestricted energy value.

Question 11: Which quantities are computed first in the feedforward direction?

A) The final loss value

B) The parameter update

C) The hidden units h_1 and h_2

D) The target label chosen by the model

Correct Answer: C

Explanation: In a feedforward network, the input layer is transformed into hidden unit values before computing the output.

Question 12: For $x_1 = 1$ and $x_2 = 1$, what is the input to the sigmoid in h_1?

A) 1

B) 0

C) 2

D) 3

Correct Answer: A

Explanation: The input to h_1 is $x_1 + x_2 - 1 = 1 + 1 - 1 = 1$.

Question 13: For $x_1 = 1$ and $x_2 = 0$, what is the input to the sigmoid in h_2?

A) -2

B) -1

C) 1

D) 2

Correct Answer: D

Explanation: The input to h_2 is $2x_1 - x_2 = 2(1) - 0 = 2$.

Question 14: Why can sigmoid hidden units introduce nonlinearity?

A) They copy inputs without changing any value.

B) They apply a nonlinear function to weighted input combinations.

C) They remove all weights from the network.

D) They force the output to be exactly linear for all possible inputs.

Correct Answer: B

Explanation: A sigmoid is nonlinear, so applying it in hidden units lets the network represent nonlinear transformations.

Question 15: What is the final output range of $\hat{y}$?

A) Any real number from $-\infty$ to ∞

B) Only the two values -1 and 1

C) A value between 0 and 1

D) A value greater than 1 for all inputs

Correct Answer: C

Explanation: The final output applies a sigmoid, so $\hat{y}$ lies in $(0, 1)$.

Case Study 4

Data for Questions 16 to 20

Warehouse Routing Time Analysis

A warehouse studies how a single routing coefficient affects the predicted travel time of a cart inside a storage aisle. The operations team first analyzes one clean training example so that the direction of a gradient descent step is easy to verify before the method is used on many routes. The one-parameter model is

$$\hat{y} = wx.$$

For one training example, $x = 3$ and $y = 15$. The squared error loss is

$$L(w) = (15 - 3w)^2.$$

The team studies this loss before using gradient descent on a larger training set. In this simplified setting, the parameter w controls how strongly the route length x affects the predicted time, and the squared loss measures how far the prediction is from the observed routing time.

Question 16: What is $L(w)$ when $w = 4$?

A) 6

B) 4

C) 12

D) 9

Correct Answer: D

Explanation: When $w = 4$, $\hat{y} = 3(4) = 12$, so $L = (15 - 12)^2 = 9$.

Question 17: Which value of w gives zero loss?

A) $w = 4$

B) $w = 5$

C) $w = 6$

D) $w = 3$

Correct Answer: B

Explanation: Zero loss requires $15 - 3w = 0$, so $w = 5$.

Question 18: What is $\frac{dL}{dw}$ at $w = 4$?

A) -18

B) -6

C) 18.0

D) 6

Correct Answer: A

Explanation: $L = (15 - 3w)^2$, so $\frac{dL}{dw} = -6(15 - 3w)$. At $w = 4$, this is -18.

Question 19: If gradient descent uses $w \leftarrow w - \eta\frac{dL}{dw}$ with $\eta = 0.1$, what is the next value from $w = 4$?

A) 3.2

B) 4.6

C) 5.8

D) 2.2

Correct Answer: C

Explanation: The update is $w = 4 - 0.1(-18) = 5.8$.

Question 20: What does the gradient indicate in this example?

A) It gives the class label directly without a model.

B) It removes the input feature from the prediction.

C) It makes the squared error impossible to compute.

D) It shows how the loss changes as w changes.

Correct Answer: D

Explanation: The gradient describes the local rate of change of the loss with respect to the parameter.

Week 3

Case Study 1

Data for Questions 1 to 5

Tutoring Session Readiness Score

An online tutoring platform uses a small feedforward network to estimate a session-readiness score before assigning a tutor. The model uses two input features:

$$x_1 : \text{student preparation score}, \quad x_2 : \text{topic difficulty score}.$$

For one session, $x_1 = 2$ and $x_2 = 1$. The network computes hidden quantities first and then combines them:

$$z_1 = 2x_1 + x_2, \qquad z_2 = x_1 - x_2 + 1,$$

$$\hat{y} = z_1 + 2z_2.$$

The computation is feedforward: inputs are used to compute z_1, z_2, and these hidden quantities are then used to compute $\hat{y}$.

Question 1: What is the value of z_1 for this session?

A) 6

B) 5

C) 4

D) 3

Correct Answer: B

Explanation: $z_1 = 2x_1 + x_2 = 2(2) + 1 = 5$.

Question 2: What is the value of z_2?

A) 0

B) 1

C) 3

D) 2

Correct Answer: D

Explanation: $z_2 = x_1 - x_2 + 1 = 2 - 1 + 1 = 2$.

Question 3: What is the final value of $\hat{y}$?

A) 9

B) 7

C) 8

D) 6

Correct Answer: A

Explanation: $\hat{y} = z_1 + 2z_2 = 5 + 2(2) = 9$.

Question 4: Which computation must be completed before $\hat{y}$ is evaluated?

A) The loss must be minimized over all stored tutoring sessions.

B) The target value must be changed by the model before scoring.

C) The hidden quantities z_1 and z_2 must be computed.

D) The learning rate must be set to zero.

Correct Answer: C

Explanation: The output depends directly on z_1 and z_2, so they must be computed first.

Question 5: What makes this a feedforward computation?

A) The output is repeatedly fed back as a new input.

B) Information flows from inputs to hidden quantities to output.

C) The target label is used before the input features.

D) The network stores a separate state for each previous session.

Correct Answer: B

Explanation: In a feedforward network, values are computed in one direction from inputs through intermediate units to the output.

Case Study 2

Data for Questions 6 to 10

Inventory Forecast Calculation Review

A store wants to understand how a change in one product-level input can affect a final forecasting loss. Before training the full inventory model, the analytics team writes a small computational graph on a whiteboard and follows both the forward values and the local derivatives. For one product, the graph is

$$u = 2x, \qquad v = u + 3, \qquad L = v^2.$$

For the current product, $x = 1$. The team computes local derivatives using the chain rule. Here, x is a simplified demand signal, u and v are intermediate quantities, and L is the scalar loss that would be reduced during training.

Question 6: What is the value of u?

A) 2

B) 1

C) 3

D) 4

Correct Answer: A

Explanation: Since $u = 2x$ and $x = 1$, $u = 2$.

Question 7: What is the value of v?

A) 3

B) 4

C) 6

D) 5

Correct Answer: D

Explanation: $v = u + 3 = 2 + 3 = 5$.

Question 8: What is the loss L?

A) 20

B) 15

C) 25

D) 10

Correct Answer: C

Explanation: $L = v^2 = 5^2 = 25$.

Question 9: What is $\frac{\partial L}{\partial v}$ at $v = 5$?

A) 5

B) 10

C) 25

D) 2

Correct Answer: B

Explanation: Since $L = v^2$, $\frac{\partial L}{\partial v} = 2v = 10$.

Question 10: What is $\frac{\partial L}{\partial x}$?

A) 10

B) 5

C) 40

D) 20

Correct Answer: D

Explanation: By the chain rule, $\frac{\partial L}{\partial x} = \frac{\partial L}{\partial v}\frac{\partial v}{\partial u}\frac{\partial u}{\partial x} = 10(1)(2) = 20$.

Case Study 3

Data for Questions 11 to 15

Sensor Calibration Error Analysis

A lab calibrates a sensor whose raw reading must be converted into a corrected measurement. To check the backpropagation calculations before using many sensor logs, the engineers use a one-hidden-unit model and one training example. For this example,

$$x = 2, \qquad y = 5.$$

The model is

$$h = w_1 x + b, \qquad \hat{y} = w_2 h,$$

with $w_1 = 1,\ b = 0,\ w_2 = 2$. The loss is

$$L = \frac{1}{2}(\hat{y} - y)^2.$$

Backpropagation computes gradients from the loss back through the output and hidden computations. The goal is not to build a production sensor model from one example, but to verify how the prediction error is assigned to w_2, then passed backward through h to w_1.

Question 11: What is the hidden value h?

A) 1

B) 4

C) 2

D) 5

Correct Answer: C

Explanation: $h = w_1 x + b = 1(2) + 0 = 2$.

Question 12: What is the prediction $\hat{y}$?

A) 3

B) 5

C) 2

D) 4

Correct Answer: D

Explanation: $\hat{y} = w_2 h = 2(2) = 4$.

Question 13: What is $\frac{\partial L}{\partial \hat{y}}$?

A) -1

B) 1

C) -2

D) 2

Correct Answer: A

Explanation: $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y = 4 - 5 = -1$.

Question 14: What is $\frac{\partial L}{\partial w_2}$?

A) -1

B) -2

C) 2

D) 1

Correct Answer: B

Explanation: Since $\hat{y} = w_2 h$, $\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} h = (-1)(2) = -2$.

Question 15: What is $\frac{\partial L}{\partial w_1}$?

A) -1

B) 1

C) -4

D) 4

Correct Answer: C

Explanation: $\frac{\partial L}{\partial h} = (-1)w_2 = -2$, and $\frac{\partial h}{\partial w_1} = x = 2$, so $\frac{\partial L}{\partial w_1} = -4$.

Case Study 4

Data for Questions 16 to 20

Course Recommendation Score Update

A course platform tracks whether a learner is likely to engage with a recommended module. For an internal training exercise, the team uses a simple linear prediction model so that each gradient and parameter update can be inspected. The model predicts a learner's interest score as

$$\hat{y} = w_1 x_1 + w_2 x_2 + b.$$

For one learner,

$$x_1 = 1, \quad x_2 = 2, \quad y = 6.$$

Initially, $w_1 = 1,\ w_2 = 1,\ b = 0$. The loss is

$$L = \frac{1}{2}(\hat{y} - y)^2.$$

The learning rate is $\eta = 0.1$, and parameters are updated using

$$w \leftarrow w - \eta \frac{\partial L}{\partial w}.$$

The two inputs summarize recent activity and topic match, while the bias accounts for a baseline interest level. The questions focus on how one observed error changes the weights and bias after a gradient descent update.

Question 16: What is the initial prediction $\hat{y}$?

A) 4

B) 3

C) 5

D) 2

Correct Answer: B

Explanation: $\hat{y} = 1(1) + 1(2) + 0 = 3$.

Question 17: What is $\frac{\partial L}{\partial \hat{y}}$?

A) -3

B) 3

C) -6

D) 6

Correct Answer: A

Explanation: $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y = 3 - 6 = -3$.

Question 18: What is $\frac{\partial L}{\partial w_2}$?

A) 3

B) -3

C) 6

D) -6

Correct Answer: D

Explanation: $\frac{\partial L}{\partial w_2} = (\hat{y} - y)x_2 = (-3)(2) = -6$.

Question 19: What is the updated value of w_1?

A) 0.7

B) 1.6

C) 1.3

D) 0.4

Correct Answer: C

Explanation: $\frac{\partial L}{\partial w_1} = (-3)(1) = -3$, so $w_1 = 1 - 0.1(-3) = 1.3$.

Question 20: What is the updated value of b?

A) 0.3

B) -0.3

C) 0.6

D) -0.6

Correct Answer: A

Explanation: $\frac{\partial L}{\partial b} = -3$, so $b = 0 - 0.1(-3) = 0.3$.

Week 4

Case Study 1

Data for Questions 1 to 5

Delivery Delay Parameter Tuning

A logistics team is tuning a simple delay model used to estimate whether a shipment route is running slower than expected. To make the first gradient descent step auditable, the team reduces the training exercise to one parameter w, where smaller loss means the parameter is closer to the best delay setting. For one training example, the loss as a function of parameter w is

$$L(w) = (w - 4)^2.$$

The team uses gradient descent:

$$w_{t+1} = w_t - \eta \frac{dL}{dw}.$$

At the current step, $w_t = 1$ and $\eta = 0.1$. The team wants to know not only the next numerical value of w, but also why the sign of the gradient determines the direction of movement.

Question 1: What is $\frac{dL}{dw}$ at $w = 1$?

A) 6

B) -3

C) 3

D) -6

Correct Answer: D

Explanation: $\frac{dL}{dw} = 2(w - 4)$. At $w = 1$, the gradient is -6.

Question 2: What is the next value of w?

A) 1.3

B) 1.6

C) 0.4

D) 2.0

Correct Answer: B

Explanation: $w_{t+1} = 1 - 0.1(-6) = 1.6$.

Question 3: Why does w increase in this update?

A) The learning rate is negative.

B) The loss is already zero.

C) The gradient at $w = 1$ is negative.

D) The update ignores the derivative.

Correct Answer: C

Explanation: Gradient descent subtracts the gradient. Subtracting a negative value increases w.

Question 4: Which value of w minimizes this loss?

A) 1

B) 2

C) 3

D) 4

Correct Answer: D

Explanation: $L(w) = (w - 4)^2$ is minimized when $w = 4$.

Question 5: What is the role of the learning rate η?

A) It controls the update step size.

B) It stores the final target value.

C) It removes the need for a gradient.

D) It makes every loss value equal.

Correct Answer: A

Explanation: The learning rate scales how far the parameter moves in the negative gradient direction.

Case Study 2

Data for Questions 6 to 10

Energy Forecast Training Schedule

A campus energy group trains a neural network to forecast short-term electricity demand for academic buildings. The full dataset is small enough for classroom discussion, but the team still wants to compare the update behavior of batch, stochastic, and mini-batch gradient descent. It has 12 training records, and in one experiment the mini-batch size is 3. One epoch means that all 12 records have been used once. The group records the gradient from each mini-batch so it can reason about the tradeoff between noisy single-record updates and more expensive full-batch updates.

Question 6: How many mini-batches are processed in one epoch?

A) 3

B) 4

C) 6

D) 12

Correct Answer: B

Explanation: With 12 records and mini-batches of size 3, the number of mini-batches is $12/3 = 4$.

Question 7: Which method uses all 12 records for each parameter update?

A) Stochastic gradient descent

B) Mini-batch gradient descent

C) Batch gradient descent

D) Momentum-only update

Correct Answer: C

Explanation: Batch gradient descent computes each update using the full training set.

Question 8: Which method uses one record per parameter update?

A) Batch gradient descent

B) Adam without gradients or single-example updates

C) Full-batch averaging

D) Stochastic gradient descent

Correct Answer: D

Explanation: Stochastic gradient descent updates parameters using one example at a time.

Question 9: If the mini-batch gradients for four batches are $2, 4, 6, 8$, what is their average?

A) 5

B) 4

C) 6

D) 8

Correct Answer: A

Explanation: The average is $(2 + 4 + 6 + 8)/4 = 5$.

Question 10: Why might mini-batch gradient descent be preferred in practice?

A) It never uses gradients.

B) It balances noisy updates and computational efficiency.

C) It requires the loss to be exactly zero before every mini-batch update.

D) It updates only after training is complete.

Correct Answer: B

Explanation: Mini-batches often provide a practical tradeoff between noisy stochastic updates and expensive full-batch updates.

Case Study 3

Data for Questions 11 to 15

Queue Prediction Parameter Update

A service desk predicts queue length so it can decide when to open additional support counters. During training, successive gradients often point in similar directions but vary in magnitude from one mini-batch to the next. To smooth the parameter movement, the team trains a model using momentum. The update is

$$v_t = \beta v_{t-1} + \eta g_t, \qquad w_t = w_{t-1} - v_t.$$

At the current step, $v_{t-1} = 0.5$, $\beta = 0.8$, $\eta = 0.1$, and the gradient is $g_t = 3$. The previous parameter is $w_{t-1} = 5$. Here, v_t stores a running update direction, so the current step depends on both the new gradient and part of the previous movement.

Question 11: What is the new velocity v_t?

A) 0.5

B) 0.6

C) 0.7

D) 0.8

Correct Answer: C

Explanation: $v_t = 0.8(0.5) + 0.1(3) = 0.4 + 0.3 = 0.7$.

Question 12: What is the updated parameter w_t?

A) 4.3

B) 5.7

C) 4.7

D) 5.3

Correct Answer: A

Explanation: $w_t = w_{t-1} - v_t = 5 - 0.7 = 4.3$.

Question 13: What does the momentum term βv_{t-1} represent?

A) It removes all history from the update and uses only the current gradient.

B) It sets the learning rate to zero for every future update.

C) It replaces the loss function.

D) It carries part of the previous update direction.

Correct Answer: D

Explanation: Momentum keeps a decayed contribution from the previous velocity.

Question 14: If $\beta = 0$, what would v_t be for the same gradient and learning rate?

A) 0.3

B) 0.1

C) 0.2

D) 0.4

Correct Answer: A

Explanation: With $\beta = 0$, $v_t = \eta g_t = 0.1(3) = 0.3$.

Question 15: Why can momentum help optimization?

A) It makes gradients unnecessary.

B) It always increases the loss.

C) It can smooth updates across steps.

D) It forces all parameters to be equal.

Correct Answer: C

Explanation: Momentum can reduce oscillations and carry useful update direction across iterations.

Case Study 4

Data for Questions 16 to 20

Demand Prediction Step-Size Analysis

A retail analytics group trains a demand prediction model whose parameters do not all receive gradients of the same scale. To understand adaptive learning rates, the team isolates one parameter and tracks how its accumulated squared gradient changes the update size. For this parameter, the current gradient is $g = 4$. In an AdaGrad-style update, the accumulated squared gradient is

$$r_t = r_{t-1} + g^2.$$

At the current step, $r_{t-1} = 9$. The learning rate is $\eta = 0.5$, and the simplified update magnitude is

$$\frac{\eta g}{\sqrt{r_t}}.$$

This simplified calculation helps the team compare a plain gradient step with an adaptive step that becomes smaller when the accumulated squared gradient is large.

Question 16: What is r_t?

A) 16

B) 25

C) 13

D) 9

Correct Answer: B

Explanation: $r_t = 9 + 4^2 = 25$.

Question 17: What is $\sqrt{r_t}$?

A) 3

B) 4

C) 5

D) 6

Correct Answer: C

Explanation: Since $r_t = 25$, $\sqrt{r_t} = 5$.

Question 18: What is the update magnitude?

A) 0.2

B) 0.4

C) 0.8

D) 0.6

Correct Answer: B

Explanation: $\eta g/\sqrt{r_t} = 0.5(4)/5 = 0.4$.

Question 19: What is the main idea behind AdaGrad-style scaling?

A) It ignores all previous gradient information while claiming to adapt steps.

B) It requires the same update for every parameter.

C) It removes the need for a loss function.

D) It adapts the step size using accumulated squared gradients.

Correct Answer: D

Explanation: AdaGrad scales updates using accumulated squared gradients, giving parameter-specific effective step sizes.

Question 20: Which optimizer combines momentum-like averaging with adaptive scaling?

A) Adam

B) Plain batch gradient descent

C) Stochastic gradient descent without memory

D) A zero-learning-rate update

Correct Answer: A

Explanation: Adam uses moving averages of gradients and squared gradients.

Week 5

Case Study 1

Data for Questions 1 to 5

Sensor Direction Scaling

A factory uses two calibrated sensors to monitor vibration and heat on a production line. During a preprocessing audit, engineers examine a linear transformation that scales the two sensor directions differently but does not mix them. The transformation is

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}.$$

The vectors $u_1 = (1, 0)^\top$ and $u_2 = (0, 1)^\top$ represent two independent sensor directions. A vector is an eigenvector of A if multiplying by A only scales the vector without changing its direction. The team wants to identify which directions keep their orientation after the transformation and how the scale factors appear as eigenvalues.

Question 1: What is Au_1?

A) $(3, 0)^\top$

B) $(0, 3)^\top$

C) $(2, 0)^\top$

D) $(1, 3)^\top$

Correct Answer: A

Explanation: $A(1, 0)^\top = (3, 0)^\top$.

Question 2: What eigenvalue corresponds to u_2?

A) 1

B) 3

C) 2

D) 0

Correct Answer: C

Explanation: $Au_2 = (0, 2)^\top = 2u_2$, so the eigenvalue is 2.

Question 3: Which statement is true about u_1?

A) It is not changed by A for any input scale.

B) It is rotated by 90 degrees after multiplication by A.

C) It becomes the zero vector.

D) It is scaled by a factor of 3.

Correct Answer: D

Explanation: $Au_1 = 3u_1$, so the direction is unchanged and the scale factor is 3.

Question 4: What is the trace of A?

A) 6

B) 5

C) 1

D) 3

Correct Answer: B

Explanation: The trace is the sum of diagonal entries: $3 + 2 = 5$.

Question 5: How are the eigenvalues related to the diagonal entries in this matrix?

A) They are equal to the diagonal entries for this diagonal matrix.

B) They must both be zero for every diagonal matrix in this setting.

C) They are found by adding all entries in each row.

D) They depend only on the off-diagonal entries here.

Correct Answer: A

Explanation: For this diagonal matrix, the eigenvalues are the diagonal entries 3 and 2.

Case Study 2

Data for Questions 6 to 10

Customer Feature Coordinate Review

A data analyst is comparing customers using two centered features: response to discounts and preference for premium products. Instead of using the original axes directly, the analyst represents each customer vector in a new basis that separates common movement from opposite movement across the two features. The basis is

$$b_1 = (1, 1)^\top, \qquad b_2 = (1, -1)^\top.$$

A customer vector is

$$x = (4, 2)^\top.$$

The analyst wants coefficients c_1, c_2 such that

$$x = c_1 b_1 + c_2 b_2.$$

Finding these coefficients shows how the same customer can be described using basis directions rather than raw feature coordinates.

Question 6: Which equation is obtained from the first coordinate?

A) $c_1 - c_2 = 4$

B) $c_1 + c_2 = 4$

C) $2c_1 = 4$

D) $c_2 = 4$

Correct Answer: B

Explanation: The first coordinate of $c_1(1, 1) + c_2(1, -1)$ is $c_1 + c_2$.

Question 7: Which equation is obtained from the second coordinate?

A) $c_1 + c_2 = 2$

B) $c_1 = 2$

C) $c_1 - c_2 = 2$

D) $2c_2 = 2$

Correct Answer: C

Explanation: The second coordinate is $c_1 - c_2$, so $c_1 - c_2 = 2$.

Question 8: What is c_1?

A) 3

B) 1

C) 4

D) 2

Correct Answer: A

Explanation: Adding $c_1 + c_2 = 4$ and $c_1 - c_2 = 2$ gives $2c_1 = 6$, so $c_1 = 3$.

Question 9: What is c_2?

A) 0

B) 1

C) 3

D) 2

Correct Answer: B

Explanation: Since $c_1 + c_2 = 4$ and $c_1 = 3$, $c_2 = 1$.

Question 10: Why do b_1 and b_2 form a basis for $\mathbb{R}^2$?

A) They point in the same direction.

B) They both have zero length.

C) They always produce only positive coordinates.

D) They are linearly independent and span $\mathbb{R}^2$.

Correct Answer: D

Explanation: Two linearly independent vectors in $\mathbb{R}^2$ form a basis.

Case Study 3

Data for Questions 11 to 15

Store Performance Feature Reduction

A retail team measures two centered features for stores: weekly footfall variation and average basket variation. The covariance matrix is

$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

PCA selects directions of maximum variance. For this diagonal covariance matrix, the coordinate axes are principal directions. The team wants to reduce the two-feature store summary to a smaller representation while retaining most of the variation across stores. Because the features have already been centered, the covariance entries describe variation around the average store behavior.

Question 11: What is the variance along the first coordinate direction?

A) 1

B) 2

C) 4

D) 0

Correct Answer: C

Explanation: The first diagonal entry of the covariance matrix is 4.

Question 12: Which direction is the first principal component?

A) The second coordinate direction

B) The direction $(1, 1)^\top$

C) The zero vector direction

D) The first coordinate direction

Correct Answer: D

Explanation: The first coordinate has the largest variance, 4, so it is the first principal component direction.

Question 13: What fraction of total variance is explained by the first principal component?

A) 4/5

B) 1/5

C) 1/4

D) 5/4

Correct Answer: A

Explanation: The total variance is $4 + 1 = 5$, and the first component explains 4.

Question 14: If the data is projected onto the first principal component only, how many dimensions remain?

A) 0

B) 1

C) 2

D) 4

Correct Answer: B

Explanation: Keeping one principal component reduces the representation to one dimension.

Question 15: What is the purpose of centering data before PCA?

A) To make every covariance value negative after centering.

B) To remove all variance from the dataset before PCA.

C) To force all eigenvalues to be equal.

D) To express variation relative to the mean.

Correct Answer: D

Explanation: Centering subtracts the mean so PCA analyzes variation around the mean.

Case Study 4

Data for Questions 16 to 20

Product-User Interaction Compression

A recommendation team studies a small product-user interaction matrix before applying matrix factorization to a larger catalog. The rows can be viewed as products and the columns as user groups, with larger entries indicating stronger interaction. The simplified matrix is

$$M = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

For this diagonal matrix, an SVD can be written as

$$M = U\Sigma V^\top$$

with $U = I$, $V = I$, and singular values 3 and 1. The team uses this small diagonal example to discuss rank, dominant singular directions, and how keeping only the largest singular value produces a low-rank approximation.

Question 16: What is the largest singular value?

A) 1

B) 2

C) 3

D) 0

Correct Answer: C

Explanation: The singular values listed are 3 and 1, so the largest is 3.

Question 17: What does $V^\top$ represent in the SVD expression?

A) The transpose of the right singular-vector matrix.

B) The loss value of the recommendation model after a full training step.

C) The number of users in every batch.

D) The covariance matrix after centering.

Correct Answer: A

Explanation: In $M = U\Sigma V^\top$, $V^\top$ is the transpose of the right singular vector matrix.

Question 18: What rank-one approximation keeps only the largest singular value?

A) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

B) $\begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$

C) $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

D) $\begin{bmatrix} 0 & 3 \\ 1 & 0 \end{bmatrix}$

Correct Answer: B

Explanation: Keeping only singular value 3 preserves the first diagonal component and drops the second.

Question 19: What is the rank of M?

A) 0

B) 1

C) 2

D) 3

Correct Answer: C

Explanation: Both diagonal entries are nonzero, so the matrix has rank 2.

Question 20: Why is SVD useful for dimensionality reduction?

A) It requires every matrix entry to be exactly zero before the factorization.

B) It removes the need for any matrix factorization.

C) It changes all singular values to negative values.

D) It can keep dominant singular directions and discard smaller ones.

Correct Answer: D

Explanation: Keeping the largest singular values gives a lower-dimensional approximation that preserves dominant structure.

Week 6

Case Study 1

Data for Questions 1 to 5

Machine Reading Compression

A maintenance team receives pairs of machine readings from an industrial unit and wants a compact summary that can still reconstruct the original reading reasonably well. To understand the mechanics of an autoencoder, the team studies one two-feature reading $x = (4, 2)^\top$ and compresses it into one latent value using an encoder

$$z = 0.5x_1 + 0.5x_2.$$

The decoder reconstructs

$$\hat{x}_1 = z, \qquad \hat{x}_2 = z.$$

The reconstruction loss is

$$L = (x_1 - \hat{x}_1)^2 + (x_2 - \hat{x}_2)^2.$$

This example lets the team separate the encoder, decoder, and reconstruction-loss roles before moving to higher-dimensional machine logs.

Question 1: What is the latent value z?

A) 3

B) 2

C) 4

D) 1

Correct Answer: A

Explanation: $z = 0.5(4) + 0.5(2) = 3$.

Question 2: What is the reconstructed vector $\hat{x}$?

A) $(2, 2)^\top$

B) $(3, 3)^\top$

C) $(4, 4)^\top$

D) $(3, 2)^\top$

Correct Answer: B

Explanation: The decoder sets both reconstructed coordinates equal to $z = 3$.

Question 3: What is the reconstruction loss?

A) 4

B) 1

C) 2

D) 0

Correct Answer: C

Explanation: $L = (4-3)^2 + (2-3)^2 = 1 + 1 = 2$.

Question 4: Which part of the autoencoder creates the latent representation?

A) The decoder output layer

B) The reconstruction loss

C) The target label table

D) The encoder layer

Correct Answer: D

Explanation: The encoder maps the input x into the latent value z.

Question 5: What is the training objective of this autoencoder?

A) To minimize reconstruction error.

B) To assign external class labels before any reconstruction is attempted.

C) To ignore the input features.

D) To make every latent value zero.

Correct Answer: A

Explanation: Autoencoders are trained to reconstruct their inputs with small error.

Case Study 2

Data for Questions 6 to 10

Report Feature Compression

A document system summarizes operational reports using five numerical features such as length, keyword density, and section-level counts. The team wants a compact representation that captures the main structure without storing all five features directly. It uses 5 input features and compresses them into a latent representation with 2 hidden units before reconstructing the 5 features. The model is undercomplete because the latent dimension is smaller than the input dimension. The setup is used to discuss why compression can force the model to learn useful structure instead of simply copying every input coordinate.

Question 6: What is the input dimension?

A) 2

B) 5

C) 7

D) 10

Correct Answer: B

Explanation: The case study states that there are 5 input features.

Question 7: What is the latent dimension?

A) 1

B) 5

C) 2

D) 7

Correct Answer: C

Explanation: The latent layer has 2 hidden units.

Question 8: Why is the model undercomplete?

A) The decoder has no reconstruction target.

B) The output layer ignores all reconstructed features during training.

C) The input has fewer features than the latent layer.

D) The latent dimension is smaller than the input dimension.

Correct Answer: D

Explanation: An undercomplete autoencoder compresses inputs into a lower-dimensional latent representation.

Question 9: Which output dimension is required to reconstruct the original input?

A) 5

B) 1

C) 2

D) 3

Correct Answer: A

Explanation: The reconstruction must have the same dimension as the 5-feature input.

Question 10: How is this related to PCA in the linear case?

A) It must always produce a one-hot class label.

B) It can learn a low-dimensional representation similar to PCA.

C) It removes the need for reconstruction in every training example.

D) It makes every component have zero variance.

Correct Answer: B

Explanation: A linear undercomplete autoencoder can learn a subspace related to PCA.

Case Study 3

Data for Questions 11 to 15

Meter Reading Recovery

A utility company receives meter readings that sometimes lose one coordinate because of temporary communication noise. To make the representation robust, it trains an autoencoder using corrupted inputs but clean targets. For one example, the clean reading is

$$x = (5, 1)^\top,$$

and the corrupted input is

$$\tilde{x} = (5, 0)^\top.$$

The model outputs $\hat{x} = (4, 1)^\top$, and the loss is

$$L = (x_1 - \hat{x}_1)^2 + (x_2 - \hat{x}_2)^2.$$

During training, the encoder sees the corrupted vector, but the loss is measured against the clean reading so the model learns to recover stable structure.

Question 11: Which vector is given to the encoder during training?

A) $x = (5, 1)^\top$

B) $\hat{x} = (4, 1)^\top$

C) $(0, 0)^\top$

D) $\tilde{x} = (5, 0)^\top$

Correct Answer: D

Explanation: A denoising autoencoder receives the corrupted input $\tilde{x}$.

Question 12: Which vector is the clean reconstruction target?

A) $(4, 1)^\top$

B) $(5, 0)^\top$

C) $(5, 1)^\top$

D) $(0, 1)^\top$

Correct Answer: C

Explanation: The clean target is $x = (5, 1)^\top$.

Question 13: What is the reconstruction loss?

A) 0

B) 2

C) 4

D) 1

Correct Answer: D

Explanation: $L = (5-4)^2 + (1-1)^2 = 1$.

Question 14: What is the purpose of using corrupted inputs?

A) To train the model to recover clean structure.

B) To remove the decoder from the network before training begins.

C) To force every output to equal zero.

D) To avoid computing any reconstruction loss.

Correct Answer: A

Explanation: Denoising autoencoders learn to reconstruct clean inputs from corrupted versions.

Question 15: Which statement best describes this training setup?

A) The model ignores all input values.

B) The model maps noisy input toward a clean target.

C) The target is the corrupted vector only.

D) The encoder output is used as the final label for every noisy example.

Correct Answer: B

Explanation: The corrupted vector is input, while the clean vector is the target.

Case Study 4

Data for Questions 16 to 20

Compact Representation Control

A research group trains an autoencoder for compact feature learning, but plain reconstruction alone produces representations that are too active and sensitive for the intended analysis. To shape the latent representation, the group starts with reconstruction loss $L_{\text{rec}} = 0.8$. It adds a sparsity penalty of 0.3 with weight $\lambda = 2$, so

$$L_{\text{sparse}} = L_{\text{rec}} + \lambda(0.3).$$

In a second experiment, the group uses a contractive penalty to make the representation less sensitive to small input changes. The two penalties illustrate different regularization goals: sparsity encourages only a few hidden units to respond strongly, while contractive regularization encourages stability.

Question 16: What is the sparsity penalty contribution?

A) 0.3

B) 0.8

C) 0.6

D) 1.1

Correct Answer: C

Explanation: The weighted penalty is $\lambda(0.3) = 2(0.3) = 0.6$.

Question 17: What is L_{sparse}?

A) 1.4

B) 1.1

C) 0.8

D) 0.6

Correct Answer: A

Explanation: $L_{\text{sparse}} = 0.8 + 0.6 = 1.4$.

Question 18: What does sparse regularization encourage?

A) Every hidden unit must always be active during the full reconstruction process.

B) The decoder is removed from the model.

C) Only a small number of hidden units should be active.

D) The input dimension must equal zero.

Correct Answer: C

Explanation: Sparse autoencoders encourage most hidden activations to be small or inactive.

Question 19: What is the goal of a contractive penalty?

A) To force reconstruction loss to be ignored throughout all training.

B) To make all weights increase without bound after every input change.

C) To replace the encoder with class labels.

D) To make the representation less sensitive to small input changes.

Correct Answer: D

Explanation: Contractive penalties encourage stable representations under small input perturbations.

Question 20: Why are such penalties added to the reconstruction loss?

A) To remove all training data from the objective.

B) To control the learned representation beyond plain reconstruction.

C) To stop the model from producing reconstructions.

D) To make every feature identical to every other feature before encoding.

Correct Answer: B

Explanation: Regularization shapes what the autoencoder learns in addition to minimizing reconstruction error.

Week 7

Case Study 1

Data for Questions 1 to 5

Water Demand Forecasting for a Residential Zone

A municipal analytics team builds models to predict next-day water demand for a residential zone. Each input contains temperature, weekday indicator, previous-day demand, and a local festival indicator. The team compares a simple linear model and a highly flexible neural network. The simple model gives similar predictions when trained on different samples, but its average prediction is far from the true demand. The neural network fits the training data very closely, but its predictions change substantially when trained on different samples.

For one input condition, the neural network predictions from four training subsets are

$$41, \quad 49, \quad 45, \quad 53$$

thousand liters, while the true demand is 48 thousand liters. The expected test error for this condition is decomposed as

$$\mathbb{E}\left[(y - \hat{f}(x))^2\right] = \text{Bias}^2 + \text{Variance} + \sigma^2.$$

Question 1: Which description best matches the simple linear model?

A) High bias and low variance

B) Low bias and high variance

C) Low bias and low variance

D) High variance and zero bias

Correct Answer: A

Explanation: Similar predictions across samples indicate low variance, while being far from the true demand indicates high bias.

Question 2: What is the average neural-network prediction for the given input?

A) 46

B) 48

C) 47

D) 50

Correct Answer: C

Explanation: The average is $(41 + 49 + 45 + 53)/4 = 188/4 = 47$.

Question 3: What is the squared bias of the neural-network predictions for this input?

A) 4

B) 9

C) 0

D) 1

Correct Answer: D

Explanation: The squared bias is $(48 - 47)^2 = 1$.

Question 4: What does the large variation among the neural-network predictions indicate?

A) The model has no irreducible error.

B) The model has high variance under the case assumptions.

C) The output labels are unavailable for the described model.

D) The model is too simple for the data.

Correct Answer: B

Explanation: Predictions changing strongly across training subsets are a symptom of high variance.

Question 5: Which action is most directly aimed at reducing the neural network's variance?

A) Add a suitable regularization method.

B) Remove the validation set entirely.

C) Increase model depth without constraints.

D) Train until training loss becomes zero.

Correct Answer: A

Explanation: Regularization methods constrain the model and help reduce overfitting-related variance.

Case Study 2

Data for Questions 6 to 10

Small-Business Loan Risk Scoring

A cooperative bank trains a logistic-regression model to predict whether a small-business loan application is risky. The model uses standardized features such as debt ratio, monthly revenue, repayment history score, and business age. Without regularization, the model reaches very high training accuracy but noticeably lower validation accuracy. Several learned weights also have large magnitudes, making the model sensitive to small feature changes.

The team adds L2 regularization to the objective:

$$J_{\text{total}}(w) = J_{\text{data}}(w) + \alpha \sum_i w_i^2.$$

For one parameter, the current value is $w = 3$. The data-loss gradient for that parameter is 4, the regularization strength is $\alpha = 2$, and the learning rate is $\eta = 0.1$. For this case, use the regularized gradient

$$\frac{\partial J_{\text{total}}}{\partial w} = 4 + 2\alpha w.$$

Question 6: What is the main purpose of adding L2 regularization here?

A) To remove the need for training data for the described model

B) To discourage large weights for the described model

C) To guarantee perfect validation accuracy

D) To convert logistic regression into clustering

Correct Answer: B

Explanation: L2 regularization penalizes large squared weights and reduces model complexity.

Question 7: What is the regularization contribution $2\alpha w$ to the gradient?

A) 10

B) 6

C) 8

D) 12

Correct Answer: D

Explanation: The contribution is $2\alpha w = 2(2)(3) = 12$.

Question 8: What is the full regularized gradient for this parameter?

A) 16

B) 12

C) 10

D) 14

Correct Answer: A

Explanation: The full gradient is $4 + 12 = 16$.

Question 9: What is the updated value of w after one gradient-descent step?

A) 2.0

B) 2.6

C) 1.4

D) 3.6

Correct Answer: C

Explanation: $w_{\text{new}} = 3 - 0.1(16) = 1.4$.

Question 10: Which outcome is expected after applying an appropriate L2 penalty?

A) Training accuracy must become zero for the data shown here.

B) Validation performance may improve due to reduced overfitting.

C) All feature weights must become exactly zero.

D) The model stops using gradient descent under the given conditions.

Correct Answer: B

Explanation: L2 regularization can reduce overfitting and improve validation performance, even if training accuracy decreases slightly.

Case Study 3

Data for Questions 11 to 15

Energy Consumption Prediction for an Office Campus

A facility-management company trains a neural network to predict hourly electricity consumption in an office campus. Inputs include outside temperature, hour of day, day type, and previous-hour load. The model is trained for many epochs, and both training and validation loss are monitored after every five epochs.

Epoch	Training Loss	Validation Loss
5	0.62	0.70
10	0.44	0.52
15	0.31	0.43
20	0.22	0.46
25	0.16	0.51

The team uses early stopping because the deployed model must generalize to future days, not only memorize historical operating patterns.

Question 11: At which epoch is the validation loss minimum?

A) Epoch 10

B) Epoch 20

C) Epoch 15

D) Epoch 25

Correct Answer: C

Explanation: The smallest validation loss is 0.43, which occurs at epoch 15.

Question 12: What pattern after epoch 15 suggests overfitting?

A) Both losses increase together.

B) Training loss increases while validation loss decreases.

C) Training loss becomes exactly zero.

D) Training loss decreases while validation loss increases.

Correct Answer: D

Explanation: After epoch 15, training loss continues decreasing but validation loss increases, indicating worsening generalization.

Question 13: Why should early stopping monitor validation loss rather than only training loss?

A) Validation loss reflects performance on unseen data.

B) Training loss is not computed during neural-network training.

C) Validation loss is always exactly equal to test loss.

D) Training loss cannot decrease across epochs.

Correct Answer: A

Explanation: Validation loss is used as a proxy for generalization to data not used for weight updates.

Question 14: Which statement about early stopping is correct?

A) It guarantees the globally optimal weights.

B) It acts as a regularization method by limiting over-training.

C) It prevents all forms of noise in input data.

D) It removes the need for a validation set under the case assumptions.

Correct Answer: B

Explanation: Early stopping limits over-training and therefore acts as a regularization method.

Question 15: If the team keeps the model from epoch 25 instead of epoch 15, what is the main risk?

A) The training set becomes larger automatically.

B) The model cannot produce numerical predictions.

C) The validation loss is ignored by definition under the case assumptions.

D) The model may generalize worse despite lower training loss.

Correct Answer: D

Explanation: Selecting epoch 25 may overfit the training data and generalize worse than the epoch with minimum validation loss.

Case Study 4

Data for Questions 16 to 20

Product-Surface Image Quality Inspection

A quality-control team trains a CNN to classify product-surface images into acceptable and defective classes. Only 900 labeled images are available. The training split is 70validation split is 30validation accuracy. The team applies image augmentation during training using small rotations, horizontal shifts, brightness changes, and mild zooms. Each original training image generates two additional augmented images.

The team also adds dropout to a hidden layer. During training, each hidden unit is kept with probability $p = 0.8$. At test time, the full network is used with appropriate scaling. Finally, three independently trained CNNs are averaged to create an ensemble.

Question 16: How many original images are in the training split before augmentation?

A) 270

B) 900

C) 630

D) 450

Correct Answer: C

Explanation: The training split contains $0.70 \times 900 = 630$ images.

Question 17: What is the effective training-set size after adding two augmented images per original training image?

A) 1890

B) 1260

C) 630

D) 1530

Correct Answer: A

Explanation: Each original contributes one original plus two augmented versions, so $630 \times 3 = 1890$.

Question 18: What is the main reason augmentation can improve validation accuracy?

A) It changes all class labels randomly in the stated case setting.

B) It removes the need to learn features.

C) It guarantees that training loss is zero.

D) It exposes the model to realistic input variations.

Correct Answer: D

Explanation: Augmentation improves robustness by exposing the model to realistic transformations of the inputs.

Question 19: During dropout training with keep probability $p = 0.8$, approximately how many units remain active in a layer of 50 hidden units?

A) 25

B) 10

C) 40

D) 50

Correct Answer: C

Explanation: The expected active units are $0.8 \times 50 = 40$.

Question 20: Why can averaging three independently trained CNNs improve reliability?

A) It increases the learning rate of each CNN.

B) It can reduce variance by combining diverse predictions.

C) It removes all bias from every CNN for the data shown here.

D) It makes the dataset labels unnecessary.

Correct Answer: B

Explanation: Averaging diverse models can cancel some individual errors and reduce prediction variance.

Week 8

Case Study 1

Data for Questions 1 to 5

Insurance Claim Approval Prediction

An insurance analytics team builds a deep feedforward network to classify whether a health insurance claim needs manual review. Each record contains numerical features such as claim amount, hospital-risk score, patient-age group, policy duration, and number of previous claims. The output layer uses a sigmoid unit because the final output is a probability.

In the first version, the team also uses sigmoid activations in all eight hidden layers. Training starts normally but soon becomes very slow. The gradients reaching early layers become very small, especially when hidden pre-activation values are large in magnitude. The team replaces the hidden activations with ReLU:

$$\mathrm{ReLU}(x) = \max(0, x),$$

while keeping the sigmoid output layer. For analysis, they compare sigmoid and ReLU outputs at $x = -4$ and $x = 2$. Use $\sigma(2) \approx 0.88$ and $\sigma(-4) \approx 0.018$.

Question 1: Why can sigmoid hidden layers slow down training in a deep network?

A) They can saturate and produce very small gradients.

B) They always output negative values under the given conditions.

C) They remove all nonlinear behavior.

D) They force every weight to become zero.

Correct Answer: A

Explanation: Sigmoid units saturate for large positive or negative inputs, making derivatives small and causing vanishing gradients.

Question 2: What is $\mathrm{ReLU}(-4)$?

A) -4

B) -1

C) 0

D) 4

Correct Answer: C

Explanation: ReLU outputs zero for negative inputs, so $\max(0, -4) = 0$.

Question 3: What is ReLU(2)?

A) 0

B) -2

C) 1

D) 2

Correct Answer: D

Explanation: For positive inputs, ReLU returns the input itself.

Question 4: Which property makes ReLU helpful for positive pre-activation values?

A) It bounds hidden outputs between 0 and 1, as sigmoid would.

B) It avoids saturation and keeps a useful gradient.

C) It converts classification into clustering.

D) It makes the loss function unnecessary.

Correct Answer: B

Explanation: ReLU has derivative 1 for positive inputs, so gradients can flow more effectively.

Question 5: Why is the output layer still kept as sigmoid in this binary classifier?

A) It maps the final score to a probability-like value between 0 and 1.

B) It is required for every hidden layer for the described model.

C) It guarantees no validation error after training on every future test case.

D) It prevents all negative input features for the data shown here.

Correct Answer: A

Explanation: A sigmoid output is suitable for binary probability prediction.

Case Study 2

Data for Questions 6 to 10

Solar Power Forecasting for a Rooftop Plant

A renewable-energy company predicts next-hour solar generation from irradiance, temperature, cloud-cover index, panel age, and previous-hour generation. The network has seven hidden layers with ReLU activations. When weights are initialized with a variance of 1, activations in later layers become unstable and the loss oscillates. The team compares Xavier and He initialization:

$$\text{Xavier: } \operatorname{Var}(w) = \frac{1}{n}, \qquad \text{He: } \operatorname{Var}(w) = \frac{2}{n},$$

where n is the number of input connections to a neuron. Consider a hidden layer with $n = 200$ inputs.

Question 6: What problem can poor initialization create in a deep network?

A) It automatically removes all noise from data.

B) It can cause activations or gradients to explode or vanish.

C) It guarantees underfitting only under the given conditions.

D) It changes regression targets into class labels.

Correct Answer: B

Explanation: Poor initialization can make signal variance grow or shrink across layers, destabilizing training.

Question 7: What is the Xavier variance for $n = 200$?

A) 0.020

B) 0.200

C) 0.100

D) 0.005

Correct Answer: D

Explanation: Xavier variance is $1/200 = 0.005$.

Question 8: What is the He variance for $n = 200$?

A) 0.010

B) 0.005

C) 0.020

D) 0.100

Correct Answer: A

Explanation: He variance is $2/200 = 0.01$.

Question 9: Why is He initialization more suitable for these hidden layers?

A) The output layer uses no activation.

B) The network has only one layer.

C) The hidden layers use ReLU activations.

D) The target variable is binary in the stated case setting.

Correct Answer: C

Explanation: He initialization includes the factor of 2 used for ReLU-style activations.

Question 10: Which observation suggests the new initialization improved training stability?

A) Hidden activations are all forced to zero.

B) Loss decreases more smoothly across epochs.

C) No validation set is needed in the stated case setting.

D) All weights become identical.

Correct Answer: B

Explanation: Smoothly decreasing loss is evidence that the optimization is more stable.

Case Study 3

Data for Questions 11 to 15

Warehouse Sensor Fault Detection

A logistics company trains a deep network to detect abnormal warehouse conditions from temperature, vibration, humidity, door-event, and power-consumption sensors. The model has ten hidden layers. Direct end-to-end supervised training is slow, and the earliest layers learn weak representations because gradients reaching them are small.

The team adopts greedy layer-wise pretraining. First, the layer closest to the input is trained with an unsupervised reconstruction objective. Its weights are then frozen, and the next layer is trained using the previous layer's representation as input. This process continues until all hidden layers are pretrained. Finally, the output layer is added and the full network is fine-tuned using the labeled abnormal-condition task.

Question 11: What is the main goal of greedy layer-wise pretraining here?

A) To delete the hidden layers before training.

B) To avoid using input features during both unsupervised pretraining and fine-tuning.

C) To give each layer a useful initialization before supervised fine-tuning.

D) To guarantee zero test error in this specific scenario.

Correct Answer: C

Explanation: Layer-wise pretraining initializes hidden layers so supervised fine-tuning starts from useful representations.

Question 12: During pretraining of the second hidden layer, what happens to the already trained first layer?

A) It is randomly reinitialized in the stated case setting.

B) It is removed from the model.

C) It is trained with the output labels only.

D) It is kept fixed while the next layer is trained.

Correct Answer: D

Explanation: Greedy layer-wise pretraining freezes earlier trained layers before training the next layer.

Question 13: Why is the first pretraining objective described as unsupervised?

A) It uses the input data without using the abnormal-condition labels.

B) It uses no numerical features under the given conditions.

C) It uses only the final classifier output for the data shown here.

D) It requires the validation labels to be hidden in the stated case setting.

Correct Answer: A

Explanation: Reconstruction-based pretraining uses inputs as training signals rather than class labels.

Question 14: What happens after all hidden layers have been pretrained?

A) The network is discarded under the given conditions.

B) The full model is fine-tuned with the supervised task objective.

C) The input features are replaced by random values.

D) The loss function is no longer required in the stated case setting.

Correct Answer: B

Explanation: After pretraining, the complete network is trained end-to-end on the labeled objective.

Question 15: Which issue is layer-wise pretraining intended to reduce?

A) The need to store data files in this specific scenario.

B) The number of output classes under the case assumptions.

C) The difficulty of training early layers in a very deep network.

D) The existence of nonlinear activations under the given conditions.

Correct Answer: C

Explanation: Pretraining helps early layers learn useful representations before full supervised fine-tuning.

Case Study 4

Data for Questions 16 to 20

Laboratory Test Result Prediction

A diagnostic laboratory predicts whether a sample needs repeat testing based on numerical features such as machine confidence score, reagent age, sample temperature, processing time, and previous calibration drift. The network has six hidden layers. Without batch normalization, training is very sensitive to learning-rate changes because layer input distributions shift during training.

The team inserts batch normalization after each linear transformation and before ReLU. For one mini-batch, a neuron's pre-activation values are

$$1, \quad 3, \quad 5, \quad 7.$$

Ignore learnable scale and shift for the numerical questions.

Question 16: What problem does batch normalization primarily address?

A) It changes labels from binary to continuous.

B) It removes the need for backpropagation.

C) It guarantees a perfect classifier in the stated case setting.

D) It reduces shifting layer-input distributions during training.

Correct Answer: D

Explanation: Batch normalization stabilizes layer input distributions across mini-batches.

Question 17: What is the mean of the mini-batch values $1, 3, 5, 7$?

A) 3

B) 4

C) 5

D) 16

Correct Answer: B

Explanation: The mean is $(1 + 3 + 5 + 7)/4 = 4$.

Question 18: What is the batch variance using the average squared deviation?

A) 4

B) 6

C) 3

D) 5

Correct Answer: D

Explanation: The squared deviations are $9, 1, 1, 9$, so the variance is $20/4 = 5$.

Question 19: Where is batch normalization placed in this case?

A) Only after the output probability for the described model.

B) Before every input feature is collected.

C) Between the linear transformation and ReLU activation.

D) After the final prediction is thresholded.

Correct Answer: C

Explanation: The case places batch normalization after the linear operation and before ReLU.

Question 20: Why are learnable scale and shift parameters useful in batch normalization?

A) They allow the network to choose an appropriate normalized representation.

B) They delete the mini-batch statistics in this specific scenario.

C) They prevent gradients from being computed under the case assumptions.

D) They force every activation to remain exactly zero for the described model.

Correct Answer: A

Explanation: The scale and shift parameters let the network learn the representation that works best after normalization.

Week 9

Case Study 1

Data for Questions 1 to 5

Customer Support Ticket Classification

A telecom company classifies customer support tickets into billing, network, plan-change, and device-support categories. After preprocessing 50,000 historical tickets, the team obtains a vocabulary of 12,000 unique words. As a baseline, every word is represented using a one-hot vector of length equal to the vocabulary size. Each vector has exactly one entry equal to 1 and all other entries equal to 0. A ticket is represented using a bag-of-words vector obtained by adding the one-hot vectors of the words present in it.

Consider two short processed tickets:

Ticket A: recharge failed refund

Ticket B: recharge plan refund

Both tickets are represented using bag-of-words vectors over the same 12,000-word vocabulary.

Question 1: What is the length of each one-hot word vector?

A) 12,000

B) 50,000

C) 3

D) 4

Correct Answer: A

Explanation: A one-hot word vector has length equal to the vocabulary size, which is 12,000.

Question 2: How many non-zero entries does a single one-hot word vector contain?

A) 12,000

B) 0

C) 1

D) 3

Correct Answer: C

Explanation: A one-hot vector has exactly one position set to 1.

Question 3: How many shared words appear in Ticket A and Ticket B?

A) 1

B) 3

C) 0

D) 2

Correct Answer: D

Explanation: The shared words are "recharge" and "refund", so there are 2 shared words.

Question 4: What is a major limitation of one-hot word vectors?

A) They are always dense for the data shown here.

B) They do not directly capture semantic similarity between words.

C) They require labels for every context window for the described model.

D) They have variable length for the same vocabulary.

Correct Answer: B

Explanation: Distinct one-hot vectors are orthogonal, so they do not encode semantic relatedness.

Question 5: Why can cosine similarity still be useful for bag-of-words ticket vectors?

A) It measures overlap in word usage while reducing sensitivity to length.

B) It learns neural hidden layers automatically from the sparse ticket vectors.

C) It converts sparse vectors into images in this specific scenario.

D) It guarantees semantic understanding under the given conditions.

Correct Answer: A

Explanation: Cosine similarity compares vector direction, which reflects word-overlap patterns in bag-of-words vectors.

Case Study 2

Data for Questions 6 to 10

Product Review Term Analysis

An online retailer studies product reviews using word co-occurrence statistics. The cleaned toy corpus contains four review fragments:

battery lasts long
screen looks bright
battery charges fast
bright screen response

The selected vocabulary is ordered as

$$[\text{battery}, \text{screen}, \text{bright}, \text{long}, \text{fast}].$$

The team uses a context window of 1, so only the immediate left and right words are counted as context. The co-occurrence matrix C has one row per target word and one column per context word.

Question 6: What is the size of the co-occurrence matrix for this selected vocabulary?

A) 4×5

B) 5×5

C) 5×4

D) 25×25

Correct Answer: B

Explanation: With five target words and the same five context words, the matrix has size 5×5.

Question 7: What does a context window of 1 mean?

A) Only the target word is counted.

B) All words in the corpus are counted as context.

C) The vocabulary is reduced to one word.

D) Only immediate neighboring words are counted.

Correct Answer: D

Explanation: Window size 1 uses the immediate left and right neighbors as context.

Question 8: In the fragment "battery lasts long", how many context contributions does "lasts" receive with window size 1?

A) 2

B) 1

C) 0

D) 3

Correct Answer: A

Explanation: The word "lasts" has "battery" on the left and "long" on the right.

Question 9: How many times are "screen" and "bright" adjacent in the toy corpus?

A) 0

B) 2

C) 1

D) 3

Correct Answer: C

Explanation: They are immediately adjacent only in "bright screen response", so the count is 1.

Question 10: What does a row of the co-occurrence matrix represent?

A) The number of labels in the classifier under the case assumptions.

B) The vectorial representation of a target word based on its context counts.

C) The one-hot position of a document category in the downstream product-review classifier.

D) The learning rate selected for truncated SVD.

Correct Answer: B

Explanation: Each row stores the target word's co-occurrence counts with context words.

Case Study 3

Data for Questions 11 to 15

Legal Search Term Representation

A legal search platform builds a word-context co-occurrence matrix from court-order summaries. The cleaned vocabulary has 6,000 words, so the square co-occurrence matrix X has size 6000×6000. The matrix is sparse and high-dimensional. To obtain compact dense word representations, the team applies truncated SVD:

$$X \approx U_k \Sigma_k V_k^\top,$$

with $k = 200$. The rows of $U_k \Sigma_k$ are used as dense word vectors for search and clustering. The platform wants related legal terms to be compared through shared context patterns rather than exact word overlap alone, which makes dimensionality reduction useful for retrieval.

Question 11: Why is truncated SVD applied here?

A) To increase the vocabulary size in the stated case setting.

B) To make every context vector one-hot.

C) To reduce dimensionality while preserving important structure.

D) To remove the need for context words in every downstream search task.

Correct Answer: C

Explanation: Truncated SVD gives compact dense vectors that retain important latent co-occurrence structure.

Question 12: What is the dimension of each reduced word vector when $k = 200$?

A) 6000

B) 36,000,000

C) 400

D) 200

Correct Answer: D

Explanation: The reduced representation has dimension $k = 200$.

Question 13: What is the size of U_k in this case?

A) 6000×200

B) 200×200

C) 200×6000

D) 6000×6000

Correct Answer: A

Explanation: U_k has one row per word and k columns, so its size is 6000×200.

Question 14: Which advantage does the reduced representation provide?

A) It makes all unrelated words identical under the given conditions.

B) It can make similarity computation more efficient.

C) It prevents any word from appearing in context.

D) It removes all latent semantic structure.

Correct Answer: B

Explanation: Comparing 200-dimensional dense vectors is more efficient than using full 6000-dimensional sparse rows.

Question 15: What kind of information can SVD help uncover?

A) The number of documents in chronological order.

B) The exact spelling of every rare word.

C) The chronological order of every document.

D) Latent patterns in word-context usage.

Correct Answer: D

Explanation: SVD can expose latent dimensions that summarize word-context usage patterns.

Case Study 4

Data for Questions 16 to 20

Search Term Similarity Evaluation

A technical support company compares two word-representation models for query expansion. For a set of word pairs, human annotators assign relatedness scores. The company computes cosine similarity between the learned word vectors and then measures how well model similarities correlate with human scores. The same vectors are also tested on a synonym selection task: given a query word and four candidates, the system chooses the candidate with the highest cosine similarity. For analogy checks of the form $a : b :: c :?$, the common search vector is $v_b - v_a + v_c$.

For one query word q, cosine similarities to four candidates are:

$$\text{A} : 0.18, \quad \text{B} : 0.74, \quad \text{C} : 0.41, \quad \text{D} : 0.62.$$

Question 16: Which candidate is selected by the synonym rule?

A) A

B) B

C) C

D) D

Correct Answer: B

Explanation: Candidate B has the largest cosine similarity, 0.74.

Question 17: What does a high correlation with human relatedness scores indicate?

A) The model's similarity judgments agree well with human judgments.

B) The vectors remain one-hot after training.

C) The vocabulary has only one word under the given conditions.

D) The model ignores cosine similarity during all human-score evaluations.

Correct Answer: A

Explanation: Higher correlation means model similarities align better with human relatedness ratings.

Question 18: In an analogy task $a : b :: c :?$, what vector operation is commonly used?

A) $v_a + v_b + v_c$ only

B) $v_a - v_b - v_c$

C) $v_b - v_a + v_c$

D) v_a / v_b

Correct Answer: C

Explanation: A common analogy query searches near $v_b - v_a + v_c$.

Question 19: Why are dense word vectors useful for query expansion?

A) They make every vector a one-hot vector under the given conditions.

B) They require no training corpus under the given conditions.

C) They make every pair of different words orthogonal.

D) They can place related words near each other in vector space.

Correct Answer: D

Explanation: Dense embeddings can encode distributional similarity, making related terms closer.

Question 20: What metric is used in the synonym rule described in the case?

A) Training loss under the given conditions

B) Matrix rank only

C) Cosine similarity

D) Vocabulary size

Correct Answer: C

Explanation: The case chooses the candidate with the highest cosine similarity.

Week 10

Case Study 1

Data for Questions 1 to 5

Warehouse Package Image Inspection

A logistics company uses a CNN to inspect package images for torn labels and damaged corners. The images are captured from a conveyor camera, and small shifts in package position should not change the final inspection decision. Each grayscale image has size 96×96. The first convolution layer applies 3×3 filters with stride 1 and padding 1 so that local edge and corner evidence can be detected without immediately changing spatial size. Later, a 2×2 max-pooling layer with stride 2 is used to reduce spatial size while retaining strong local responses.

Question 1: What is the output spatial size after the 3×3 convolution with stride 1 and padding 1?

A) 96×96

B) 94×94

C) 98×98

D) 48×48

Correct Answer: A

Explanation: $(96 + 2(1) - 3)/1 + 1 = 96$, so the spatial size is preserved.

Question 2: If the convolution layer uses 24 filters, how many output feature maps are produced?

A) 3

B) 1

C) 24

D) 96

Correct Answer: C

Explanation: Each convolution filter produces one feature map.

Question 3: What is the output size after applying 2×2 max pooling with stride 2 to a 96×96 feature map?

A) 96×96

B) 24×24

C) 94×94

D) 48×48

Correct Answer: D

Explanation: Pooling with size 2 and stride 2 halves both height and width.

Question 4: Why is pooling useful for this inspection task?

A) It removes the need for convolution under the case assumptions.

B) It reduces spatial size and gives some robustness to small shifts.

C) It increases every image dimension in the stated case setting.

D) It converts grayscale images into RGB images in this specific scenario.

Correct Answer: B

Explanation: Pooling reduces resolution while preserving strong local evidence.

Question 5: What do early convolution filters usually learn in such image tasks?

A) Low-level patterns such as edges and corners

B) The final class probability only

C) The batch size for the described model

D) The optimizer momentum in this specific scenario

Correct Answer: A

Explanation: Early CNN filters commonly detect simple visual patterns such as edges, corners, and textures.

Case Study 2

Data for Questions 6 to 10

Crop Image Disease Detection

An agronomy lab trains a CNN to classify crop leaf images as healthy or diseased. Researchers need to verify that the network focuses on leaf lesions rather than background soil. They use occlusion analysis by masking small patches and observing the change in disease probability. They also inspect activation maps from early and deeper layers. This review is important because a high validation score alone would not show whether the model is using biologically meaningful lesion patterns. The lab therefore treats visualization as a diagnostic tool for understanding which regions and filters influence the prediction.

Question 6: If masking one leaf patch sharply reduces disease probability, what does that suggest?

A) The patch is irrelevant to the prediction and all class evidence.

B) The patch contains information important for the prediction.

C) The model has no learned filters for the data shown here.

D) The image has no disease signal for the data shown here.

Correct Answer: B

Explanation: A large confidence drop means the model relied strongly on that region.

Question 7: What is the name of the technique that masks image regions and observes prediction changes?

A) Batch normalization

B) Weight decay

C) Dropout

D) Occlusion analysis

Correct Answer: D

Explanation: Occlusion analysis studies sensitivity by hiding parts of the input.

Question 8: Why are first-layer filters usually easier to interpret?

A) They operate directly on local pixel patterns.

B) They contain no weights in this specific scenario.

C) They always output class labels.

D) They are identical across all filters.

Correct Answer: A

Explanation: First-layer filters respond to visible pixel-level features such as edges and textures.

Question 9: A neuron that activates strongly for diseased leaves but weakly for healthy leaves has likely learned what?

A) The file name pattern in this specific scenario

B) A disease-related visual feature

C) The learning rate schedule

D) A fixed image size only

Correct Answer: B

Explanation: Class-selective activation suggests the neuron detects a disease-relevant pattern.

Question 10: In gradient-based visualization, what does a large gradient magnitude for a pixel indicate?

A) The pixel has no influence for the data shown here.

B) The pixel is outside the image.

C) The activation is sensitive to that pixel.

D) The filter has zero parameters.

Correct Answer: C

Explanation: Large gradients indicate that small pixel changes can strongly affect the activation.

Case Study 3

Data for Questions 11 to 15

Road Surface Image Classification

A transport analytics team classifies road-surface images into smooth, cracked, pothole, and waterlogged categories. They compare VGG-style stacks of 3×3 filters, Inception-style parallel filters, and ResNet-style skip connections. The images contain both small cracks and larger road patterns, so features at multiple spatial scales are important. The team is choosing an architecture for a field system that must detect fine texture changes without losing larger context such as standing water or pothole boundaries. The comparison focuses on how different CNN designs build depth, combine filter sizes, and preserve gradient flow during training.

Question 11: Why does VGG prefer stacks of small 3×3 filters?

A) To remove nonlinearities in the stated case setting.

B) To avoid feature maps throughout the network.

C) To force all images to be grayscale before every convolutional layer.

D) To build deeper representations with controlled parameter growth.

Correct Answer: D

Explanation: Stacked small filters increase effective receptive field while keeping parameters manageable.

Question 12: What is the main purpose of Inception-style parallel filters?

A) To delete pooling layers.

B) To use only one spatial scale.

C) To capture features at multiple spatial scales.

D) To remove convolution in this specific scenario.

Correct Answer: C

Explanation: Parallel filters of different sizes detect fine and coarse features in the same module.

Question 13: Why do ResNet skip connections help very deep networks train?

A) They improve information and gradient flow.

B) They remove the input image.

C) They replace all filters with one scalar.

D) They make the output size independent of stride.

Correct Answer: A

Explanation: Skip connections provide direct paths that ease optimization in deep networks.

Question 14: A module computes $H(x) = F(x) + x$. What architecture idea does this represent?

A) Bag-of-words

B) Residual learning

C) One-hot encoding

D) Average pooling only

Correct Answer: B

Explanation: Residual blocks add the input x to the learned residual function $F(x)$.

Question 15: Which architecture is most directly suited to mixing 1×1, 3×3, and 5×5 convolutions in parallel?

A) Inception in this specific scenario

B) A single-layer perceptron

C) Plain linear regression

D) A one-hot encoder for the described model

Correct Answer: A

Explanation: Inception modules use parallel convolution branches with different filter sizes.

Case Study 4

Data for Questions 16 to 20

Satellite Land-Cover Image Classification

A satellite-imaging team classifies 64×64 RGB patches into urban, forest, water, and farmland. One convolution layer uses 32 filters, each of size 5×5, with stride 1 and no padding. Another stage uses global average pooling to reduce a final $8 \times 8 \times 128$ feature tensor into a 128-dimensional vector before classification. The analysts need to estimate both spatial output sizes and parameter counts before scaling the model to many satellite tiles. They also want to understand why global average pooling can summarize feature maps without introducing a very large dense layer.

Question 16: What is the output spatial size after the 5×5, stride 1, no-padding convolution on a 64×64 image?

A) 64×64

B) 32×32

C) 60×60

D) 68×68

Correct Answer: C

Explanation: $(64 - 5)/1 + 1 = 60$.

Question 17: How many output channels are produced by 32 filters?

A) 5

B) 32

C) 64

D) 128

Correct Answer: B

Explanation: The output depth equals the number of filters.

Question 18: How many weight parameters are in one 5×5 RGB filter, ignoring bias?

A) 25

B) 50

C) 125

D) 75

Correct Answer: D

Explanation: One filter spans all 3 input channels, so $5 \times 5 \times 3 = 75$.

Question 19: What is the output dimension after global average pooling of an $8 \times 8 \times 128$ tensor?

A) 64

B) 1024

C) 128

D) 8192

Correct Answer: C

Explanation: Global average pooling averages each of the 128 feature maps to one value.

Question 20: Why can global average pooling reduce parameters compared with flattening followed by a large fully connected layer?

A) It increases the spatial dimensions.

B) It removes all learned filters in this specific scenario.

C) It increases the number of dense connections.

D) It produces one summary value per feature map.

Correct Answer: D

Explanation: Global average pooling avoids connecting every spatial position to a large dense layer.